

MTL106: Probability and Stochastic Processes

(Practice Sheet 1)

1. Write the sample spaces for the following cases:

- (a) A company is launching a marketing campaign and wants to test the three different advertisements (A, B, and C) on two different social media platforms (Facebook and Instagram).
- (b) In a tennis tournament, a player can win (W) or lose (L) each match. The event is recording the outcomes of the player's next three matches.
- (c) A weather station predicts the weather for the next two days, with each day being either sunny (S), rainy (R), or cloudy (C).

2. Let the sample space $\Omega = \{1, 2, 3, 4\}$. Find three different σ -fields, $\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3$ such that

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3.$$

3. Let the sample space $\Omega = \{1, 2, 3\}$, where each element represents a distinct and mutually exclusive outcome, and $\mathcal{F}$ is the σ -field (consider power set) on Ω . Let us define P on $\mathcal{F}$ as follows:

$$P(1) = 0.2, \quad P(2) = 0.5, \quad P(3) = 0.3.$$

Check whether P is a probability function or not.

4. Let the sample space $\Omega = [0, 1]$, and $\mathcal{F}$ is the Borel σ -field on Ω . Let us define P on $\mathcal{F}$ as follows:

$$P = \begin{cases} 2x, & x \in [0, 1/2], \\ 2(1-x), & x \in (1/2, 1]. \end{cases}$$

Check whether P is a probability function or not.

5. A website displays 10 advertisements and the revenue generated by the website depends on the number of visitors to the site clicking on any of the advertisements displayed on the website. The data collected by the company has revealed that out of 2500 visitors, 30 visitors clicked on 1 advertisement, 15 clicked on 2 advertisements, and 5 clicked on 3 advertisements. Remaining did not click on any of the advertisements. Calculate

- (a) The probability that a visitor to the website will click on an advertisement.
- (b) The probability that the visitor will click on at least two advertisements.
- (c) The probability that a visitor will not click on any advertisements.

6. What is the probability that a randomly selected leap year will contain 53 Sundays?

7. For any two events A and B , prove that

$$P\{(A \cap B^c) \cup (A^c \cap B)\} = P(A) + P(B) - 2P(A \cap B).$$

8. There are 280 students in the class of MTL106 course at IIT Delhi. What is the probability that all these students have different birthdays (in a year of 365 days)?

9. In the above example (Q8), what is the probability that at least two students (out of 280) have same birthdays?

10. A room has three electric lamps. From a collection of 10 electric bulbs of which 6 are good, 3 are selected at random and put in the lamps. Find the probability that the room is lighted.